

Project Scope

Project Title

AI-Powered Credit Risk Analytics & Loan Default Prediction Platform

Project Overview

This project aims to analyze historical customer, loan, credit bureau, and repayment data to identify risk patterns, predict loan defaults, and support data-driven lending decisions. The solution combines data analytics, machine learning, business intelligence, and AI to build a comprehensive credit risk assessment framework.

Scope Statement

The project focuses on understanding borrower behavior, identifying key risk factors, predicting loan default probability, and providing actionable business insights. The final solution will serve as a decision-support system for credit risk management and lending operations.

In-Scope Activities

Business Understanding

- Business problem analysis
- Stakeholder analysis
- Credit risk domain understanding
- KPI and success metric definition

Data Analytics

- Data profiling and quality assessment
- Data cleaning and transformation
- Exploratory Data Analysis (EDA)
- Statistical analysis

SQL Analytics

- Customer segmentation
- Credit risk analysis
- Portfolio performance analysis
- Business query development

Machine Learning

- Feature engineering
- Default prediction models
- Model evaluation and comparison
- Explainable AI analysis

Business Intelligence

- Power BI dashboard development
- KPI monitoring and reporting
- Interactive risk analysis

AI Integration

- AI-powered risk explanations
- Lending recommendation generation
- Customer risk summaries

Key Deliverables

Business Deliverables

- Credit Risk Framework
- Customer Risk Segmentation
- Business Recommendations

Technical Deliverables

- SQL Analysis
- Python EDA Notebook
- Statistical Analysis Report
- Machine Learning Models
- Power BI Dashboard
- AI Risk Assistant
- Streamlit Application

Documentation Deliverables

- Business Understanding Documents
- Domain Knowledge Repository
- Data Dictionary
- Final Executive Report
- GitHub Repository

Out of Scope

The following areas are excluded from this project:

- Fraud Detection
- Anti-Money Laundering (AML)
- Real-Time Loan Approval Systems
- Production Deployment
- Cloud Infrastructure Setup
- API Development
- Deep Learning Models
- Regulatory Certification

Data Scope

The project will utilize the Home Credit Default Risk Dataset, including:

- Application Data
- Credit Bureau Data
- Previous Loan Applications
- Repayment History
- Credit Card Information
- POS and Cash Loan Records

The analysis will focus on historical lending behavior and customer credit profiles.

Technology Scope

Area	Tools
Data Analysis	SQL, Python
Statistics	SciPy, Statsmodels
Machine Learning	Scikit-learn, XGBoost
Visualization	Matplotlib, Seaborn, Plotly
Business Intelligence	Power BI
AI Integration	Gemini
Application Development	Streamlit
Version Control	GitHub

Success Criteria

The project will be considered successful if it:

- Identifies key drivers of loan default.
- Accurately predicts high-risk borrowers.
- Provides actionable business recommendations.
- Delivers an interactive Power BI dashboard.
- Demonstrate SQL, Python, Machine Learning, and AI capabilities in a single end-to-end project.

Conclusion

This project covers the complete credit risk analytics lifecycle, from business understanding and data preparation to predictive modeling, dashboard development, and AI-assisted decision support. The scope is designed to deliver meaningful business insights while showcasing advanced analytics skills through a real-world financial services use case.